BrickBuddy

AI-Powered LEGO Robot Building Assistant for Children (6-8) | Yusong HUANG, Jiayi HUANG | CMAA5043 Milestone Check-in

Project Goal

- Multimodal AI assistant guiding kids through LEGO robot building
- Voice + camera + template input for natural child expression
- Personalized step-by-step instructions with emotional support
- Integrated STEAM education throughout the building process

3-Stage Interaction Flow

- **Stage 1 - Imagine:** Child describes/shows robot idea via voice, camera, or templates
- **Stage 2 - Build:** AI generates adaptive instructions + real-time chat support
- **Stage 3 - Learn:** STEAM Q&A summarizes science, tech, engineering, art, math concepts

STEAM Integration

Science | Technology | Engineering | Art | Math

Interface Design (5 Screens)

- **Splash:** Welcome mascot + "Let's Build!" CTA
- **Imagine:** Voice/camera/template multimodal input
- **Build:** Step-by-step instructions + AI chat sidebar
- **Learn:** STEAM Q&A cards with fun facts
- **Celebrate:** Confetti + progress stats + learning summary

Progress

Done	Done	Next
Hi-Fi Prototype	5 Screens	User Testing

What's Next

- Integrate real voice recognition (Web Speech API)
- Connect camera for real-time piece detection
- User testing with children aged 6-8
- Expand LEGO model library beyond Robot Dog

Live Demo: yusong-enceladus.github.io/CMAA5043/week9/prototype.html